

Mehmet Arif Demirtas

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Summary

I'm **Arif** (he/him)! In my research, I develop and evaluate systems to support instructors and students through human-centered design. I focus on computing education, specifically for students who are **not** majoring in Computer Science.

Education

University of Illinois Urbana-Champaign

PhD, Computer Science

Aug 2023 – May 2028

(expected)

- **Advisor:** Dr. Katie Cunningham

MS, Computer Science

Jan 2026

- **Thesis:** *Developing and evaluating domain models for programming skills using learning curve analysis*

Istanbul Technical University

Sep 2018 - Jan 2023

BS, Computer Engineering

- **Thesis:** *Automated Realistic Lip Sync Generation for Unconstrained Videos*

Teaching

Lead Teaching Assistant, **Modeling and Learning in Data Science**, UIUC

Spring 2026

Teaching Assistant, **Modeling and Learning in Data Science**, UIUC

Fall 2025

- Leading hour-long discussion sections for CS307 - Modeling and Learning in Data Science with Dr. David Dalpiaz (Fall 2025) and Dr. Pablo Robles Granda and Dr. Dan Gonzalez (Spring 2026)
- Preparing and teaching lectures on ML fundamentals and applications with Scikit-Learn for over 100 students

Instructor / Facilitator, **Machine Learning Foundations**, Break Through Tech

Summer 2026

Course Support, **Machine Learning Foundations**, Break Through Tech

Summer 2025

- Supported the small group discussions in the lectures for the 9-week ML Foundations course with 52 students
- Provided just-in-time instruction for weekly assignments

Lecturer, **Introduction to Python**, ITU ACM Student Chapter

Fall 2022

- Held introductory Python lectures and supervised hands-on tutorial sessions for more than 100 students

Guide & Mentor, **Deep Learning Study Group**, inzva.com

Oct 2021 - Nov 2022

- Gave lectures on object detection, face recognition, and neural style transfer to more than 100 students as part of the Google Developers ML Bootcamp and Deep Learning Study Group at inzva hackerspace

Selected Publications

* denotes equal contribution.

- [FP6] **M. A. Demirtas**, S. Nam, and G. Griffin. “Leveraging Human-AI Collaboration for a Passage-Based Question Authoring Tool”. In: *International Conference on Artificial Intelligence in Education*. Springer Nature Switzerland. 2026.
- [FP5] **M. A. Demirtas**, C. Zheng, and K. Cunningham. “Teaching Authentic Programming Applications to Novices: Purpose-first Tutorials in a General Education Computing Course”. In: *Proceedings of the 57th ACM Technical Symposium on Computer Science Education V.1*. New York, NY, USA: Association for Computing Machinery, 2026, pp. 274–280.
- [FP4] Y. Jain*, **M. A. Demirtas***, and K. Cunningham. “PLAID: Supporting Computing Instructors to Identify Domain-Specific Programming Plans at Scale”. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 2025, pp. 1–21.

- [FP3] **M. A. Demirtaş**, C. Zheng, M. Fowler, and K. Cunningham. “Generating Planning Feedback for Open-Ended Programming Exercises with LLMs”. In: *International Conference on Artificial Intelligence in Education*. Springer Nature Switzerland. 2025, pp. 33–48.
- [FP2] **M. A. Demirtaş**, M. Fowler, and K. Cunningham. “Reexamining Learning Curve Analysis in Programming Education: The Value of Many Small Problems”. In: *Proceedings of the 17th International Conference on Educational Data Mining*. International Educational Data Mining Society. 2024, pp. 53–67.
- [FP1] **M. A. Demirtaş**, M. Fowler, N. Hu, and K. Cunningham. “Validating, Refining, and Identifying Programming Plans Using Learning Curve Analysis on Code Writing Data”. In: *Proceedings of the 2024 ACM Conference on International Computing Education Research-Volume 1*. 2024, pp. 263–279.

Short Papers

- [SP3] **M. A. Demirtaş**, J. Hur, A. Atcheson, C. Zheng, and K. Cunningham. “Towards Supporting Novice Programmers’ Web Search Skills for Using Unfamiliar Libraries and APIs”. In: *Proceedings of the 37th Annual Workshop of the Psychology of Programming Interest Group*. 2026.
- [SP2] R. Sampaio de Alencar, **M. A. Demirtaş**, A. S. Saha, Y. Shi, and P. Brusilovsky. “Integrating Expert Knowledge With Automated Knowledge Component Extraction for Student Modeling”. In: *Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization*. 2025, pp. 307–312.
- [SP1] S. Yürekli, **M. A. Demirtaş**, and I. Rekik. “Quantifying the Predictive Uncertainty of Regression GNN Models Under Target Domain Shifts”. In: *Predictive Intelligence in Medicine: 5th International Workshop, PRIME 2022, Held in Conjunction with MICCAI 2022, Singapore, September 22, 2022, Proceedings*. Springer Nature Switzerland Cham. 2022, pp. 149–159.

Posters & Extended Abstracts

- [PEA3] **M. A. Demirtaş**, C. Zheng, and K. Cunningham. “Detecting Programming Plans in Open-ended Code Submissions”. In: *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 2*. 2025, pp. 1435–1436.
- [PEA2] **M. A. Demirtaş**. “Identifying and Evaluating Novel Knowledge Component Models for Programming Skills”. In: *Proceedings of the 17th International Conference on Educational Data Mining*. Ed. by B. Paassen and C. D. Epp. Atlanta, Georgia, USA: International Educational Data Mining Society, July 2024, pp. 969–973. ISBN: 978-1-7336736-5-5. DOI: 10.5281/zenodo.12730015 [↗](#).
- [PEA1] E. S. Wiese, J. Finnie-Ansley, R. Duran, K. Cunningham, and **M. A. Demirtaş**. “Challenges and Solutions for Teaching Decomposition and Planning Skills in CS1”. In: *Proceedings of the 2024 on ACM Virtual Global Computing Education Conference V. 2*. 2024, pp. 291–292.

Selected Professional Experience

AI/ML Research Intern

ACT Inc.

Iowa City, USA (Remote)

June 2025 – Aug 2025

- Led a human-centered design project for improving question authoring experience for subject matter experts creating multiple-choice reading comprehension questions for the ACT exam
- Conducted interviews to identify design opportunities for authoring tools with subject matter experts
- Designed and implemented a human-AI collaboration tool for question authors to utilize LLM-generated suggestions in early-stage ideation processes to improve user experience and productivity
- Evaluated the impact of the tool in a mixed-methods user study, presented at **AIED 2026** [FP6]

Undergraduate Mentorship

Claire Zheng, Fall 2024 - Spring 2026

- Contributed to [FP3, FP5], responsible for data processing and analysis, writing, and pilot studies
- Next position: SWE @ Google

Yoshee Jain, Spring 2024 - Fall 2024

- Collaborated on [FP4] as co-first author

- Next position: PhD Student @ UIUC

Nicole Hu, Spring 2024

- Contributed to [FP1], responsible for data processing and annotation

Honors & Awards

UIUC Facilitating Learning Excellence Award, 2026.

Future Leaders of AI Travel Scholarship, ACM AI Summit (\$1250), 2026.

Doctoral Consortium Travel Scholarship, EDM 2024 (\$2500), 2024.

Grants

Building a community of practice to support programming education across engineering departments. Strategic Instructional Innovations Program, The Grainger College of Engineering (\$6500). 2026.

- **PIs:** Katie Cunningham, Ali Ansari, Bryan K Clark, Christopher Tessum, Elizabeth Wickes, Max Fowler, **Mehmet Arif Demirtas**.
- **Role:** Co-author of proposal and corresponding co-PI for research activities.

Service

Panel Moderator, SIGCSE Virtual 2024

- *Challenges and Solutions for Teaching Decomposition and Planning Skills in CS1* with Dr. Eliane S Wiese, James Finnie-Ansley, Dr. Rodrigo Duran, Dr. Katie Cunningham

Reviewer: SIGCSE TS 2027-2026-2025, CHI 2025, SIGCSE Virtual 2024, Computer Science Education

Student Research Competition Judge: ISUR Research Expo 2026